From Person-Specific Dynamics to Population Inference: A Computationally Efficient Meta-Analytic Structural Equation Modeling Framework for Integrating Complex Multilevel Dynamic Models

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Intensive longitudinal research often seeks population-level inference on within-person dynamics without sacrificing idiographic estimation. We introduce MetaVAR, a two-stage meta-analytic structural equation modeling (MASEM) framework for multilevel dynamic systems that combines person-specific dynamic modeling with multivariate meta-analytic synthesis. In Stage 1, person-specific vector autoregressive (VAR) models are fit to obtain each individual's dynamic parameter vector and corresponding sampling covariance matrix. In Stage 2, those parameter vectors are synthesized using a multivariate random-effects meta-analytic SEM, allowing population-level inference on average dynamics, between-person heterogeneity, and cross-parameter covariation while propagating first-stage uncertainty. In a Monte Carlo simulation, we compare MetaVAR with (a) a one-step Bayesian multilevel VAR benchmark implemented in `Mplus` under default or relatively uninformative priors and (b) an uncertainty-uncorrected two-stage fit-many-then-summarize benchmark. Results indicate that MetaVAR's clearest advantage lies in recovery of between-person heterogeneity and calibration of interval estimates for heterogeneity parameters, whereas the one-step Bayesian benchmark remains competitive for population-average effects. An ecological momentary assessment illustration on negative and positive affect further shows that MetaVAR recovers interpretable average dynamics together with substantial heterogeneity in affective set points, inertia, cross-lagged coupling, and innovation structure.

Keywords: vector autoregressive model, multilevel model, random-effects meta-analysis, structural equation modeling

As a prior-sensitivity analysis, we refit the Bayesian multilevel VAR using a common set of relatively weakly informative, scale-aware priors held constant across all heterogeneity conditions. Population-average set points received $\mathcal{N}(0, 25)$ priors, whereas population-average transition coefficients received $\mathcal{N}(0, 0.25)$ priors. The corresponding prior standard deviations of 5 and 0.5 were broad relative to the respective parameter scales. The within-person innovation covariance and between-person set-point covariance matrices received $IW(\mathbf{I}_2, 3)$ priors. Because transition-coefficient heterogeneity was expected to occur on a substantially smaller scale, its covariance matrix received an $IW(0.004\mathbf{I}_4, 5)$ prior. With four random transition coefficients, this specification implies marginal $IG(1, 0.002)$ priors for the transition variances, with mode 0.001 and a heavy upper tail. Inverse-Wishart prior calibration is consequential when random-effect variances are small or near the boundary; we therefore compared results under these priors with those obtained using the Mplus default priors (Muthén & Muthén, 2017; Schuurman et al., 2016; van Erp et al., 2018).

For the four-variable feasibility analysis, we conducted a prior-sensitivity analysis for the reference simulation condition, with (n = 200), (T = 200), and the reference level of heterogeneity. Population-average set points received $\mathcal{N}(0, 25)$ priors, whereas population-average transition coefficients received $\mathcal{N}(0, 0.25)$ priors. The corresponding prior standard deviations of (5) and (0.5), respectively, were broad relative to the parameter scales. The four-dimensional within-person innovation covariance matrix received an $IW(\mathbf{I}4, 5)$ prior. In the Mplus implementation, the 10 unique elements of the lower triangular covariance matrix were assigned inverse-Wishart scale values element by element, with unit values for the four diagonal elements and zero values for the six off-diagonal elements, thereby defining **I**4 as the prior scale matrix. Because the between-person covariance structures were diagonal in the four-variable model, independent $IG(2, 0.025)$ priors were assigned to the set-point and transition-coefficient variances. Under the Mplus shape–scale parameterization, this inverse-gamma prior has mean (0.025), mode (0.0083), and a heavy upper tail. The prior was calibrated to the order of magnitude of the random-effect variances specified in the reference simulation design, rather than to estimates obtained from individual replications, and was held constant across replications. We compared results under these scale-aware priors with those obtained using the Mplus default priors. This comparison follows recommendations to evaluate prior sensitivity explicitly because apparently diffuse covariance priors can be consequential when random-effect variances are small or near the boundary (Muthén & Muthén, 2017; Schuurman et al., 2016; van Erp et al., 2018).

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